

Identifying the spatiotemporal dynamics of forest ecotourism values with remotely sensed images and social media data: A perspective of public preferences

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ABSTRACT

Ecotourism, as a cultural ecosystem service, is an essential part of rural economic growth. Developing feasible methods and indicators for ecotourism values is vital to exploring the interactions between public preferences and ecosystems, thus providing information for sustainable ecosystem management. However, forest ecotourism value assessments at the regional scale still lack scientific frameworks covering the spatiotemporal dimension. Here, we proposed a framework to assess forest ecotourism values and explore the contributions of public preferences, selecting a typical and abundant forest area, Lin'an County, China. First, we applied a text mining approach on social comments to identify key variables related to public preferences. Then, we mapped these variables with remote sensing and GIS data and incorporated them with a Maxent model. This model revealed the spatiotemporal patterns of forest ecotourism value and its interaction with public preferences, over two periods, i.e., 2010–2014 and 2015–2019. Our results showed that albeit with a minor change magnitude in the entirety of ecotourism values (1–3% overall decrease), about 34% and 44% of forests were observed as significant increases and decreases in ecotourism values, respectively. A large number of increases were concentrated in the village scenic areas, while decreases were observed in planted areas. Such notable changes were mainly attributable to changes in accessibility variables with 67% of contributions in 2010–2014 and 72% in 2015–2019. Variables related to human perception and vision (forest attribute and color diversity) also played a part. Our findings infer that local rural-building policies have an evident influence on the distribution and intensity of forest ecotourism. Our proposed framework provides references for evaluating the response and effectiveness of local policies and to make proactive efforts for forest protection, restoration and management.

1. Introduction

Forests play a paramount role in terrestrial ecosystems, supporting livelihood, conserving biodiversity, providing fresh air and clean water, as well as mitigating climate change (FAO, 2020). They contribute to human well-being by offering material benefits (i.e., provisioning and regulating ecosystem services) and valuable non-material benefits (i.e., cultural ecosystem services) (MEA, 2005). Among them, forest ecotourism, defined as the non-material benefits provided by forest

ecosystems through spiritual enrichment, recreation, and aesthetic experiences (Havanga et al., 2020; MEA, 2005), is gradually becoming an essential contributor to economic growth in mountainous areas. However, forests ecotourism values are highly susceptible to human activities (Yu et al., 2019), resulting in significant variations in both spatial and temporal dimensions. Understanding the spatiotemporal dynamics of forest ecotourism value and its interaction with public preferences is vital to sustainable forest development and rural revitalization (Knoke et al., 2021; Plieninger et al., 2015).

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Poverty and biodiversity hotspots are geographically overlapped and clustered in rural areas where livelihoods are disproportionately propped up by ecological products and ecosystem services in forests (Barrett et al., 2011). Decision makers normally formulate strategic policies to stimulate the economy. In China, the total output value of forestry industries has sharply increased by 420% from 1978 (17.96 billion yuan) to 2019 (7.56 trillion yuan), expected sustained rapid growth (NFGA, 2019). As of 2018, more than 465 thousand poor households relied on forest tourism to increase their income (NFGA, 2019). The demands for landscape aesthetics and forest recreation were mounting up, especially for people who live in cities far away from nature. Another reason for growing visitor numbers in mountainous areas is improved accessibility as a result of the construction of related tourism infrastructure (Schirpke et al., 2019). In such mountainous or rural areas, economic development has received much attention, while the processes of ecosystems are poorly understood. These increasing pressures require decision makers to place greater emphasis on forest ecosystem management so as to meet the diverse needs of visitors as well as to protect some ecologically sensitive areas.

The assessments of ecotourism values in mountainous areas have been continually conducted. Traditional assessments based on empirical (questionnaire surveys, etc.) and participatory (expert knowledge, etc.) methods are usually expensive, time-consuming, and limited in spatio-temporal dimensions (e.g., Giergiczny et al., 2015; Hegetschweiler et al., 2017; Schirpke et al., 2021). Current methods of assessments using spatialized indicators encounter difficulties in capturing the subjectivity of visitors (e.g., Hernández-Morcillo et al., 2013; Mengist et al., 2020). The information collected by the survey or the setting of the indicator framework is often both subjective and biased. For instance, partial forest characteristics, the number of large trees and the amount of dead-wood, support forest assessments in terms of ecotourism, but are rarely used (Blicharska and Mikusiński, 2014; Knoke et al., 2021). Besides, the impact of policy implementation poses challenges for identifying ecotourism values and their changes (Fish et al., 2016). Unable to present the spatial heterogeneity of ecotourism values or temporal change is always ignored (Kim et al., 2019).

Integrated data from social media and earth observation provides viable and promising spatial-temporal substances for understanding public preferences on ecotourism values. Recently, social networks offer visitors online channels to share photos and express opinions. These geotagged photos and relevant comments can be freely acquired (Van Zanten et al., 2016). Photo content analysis for measuring landscape features provides an effective perspective to assess ecotourism values (e.g., Ciesielski and Stereńczak, 2021; Yoshimura and Hiura, 2017). However, these platforms (e.g., Flickr, Instagram) are not popular in China. Mining of public preferences in social comments also remains understudied (Gosal et al., 2019; Li et al., 2021). The extensive location information updated through the interaction between the platform and its users is an important complement to traditional ground surveys (He et al., 2019). Moreover, remote sensing (RS) data can directly extract visual and sensory attributes, such as vegetation coverage and landscape patterns, and their changes at the regional scale (Cord et al., 2017; Vaz et al., 2019). Especially, the free-to-use Google Earth Engine (GEE) platform offers powerful computing capabilities to analyze such geospatial big data (Gorelick et al., 2017). In short, establishing a comprehensive, targeted and effective predictive model for spatial and temporal forest ecotourism values remains a challenging task.

This study aims to propose a feasible and scientific framework to understand the spatiotemporal dynamics of forest ecotourism value and associated public preferences in a typical mountainous area of Lin'an County, China during 2010–2019. The specific goals were to (1) identify key variables related to public preferences; (2) quantify the distributions of forest ecotourism values and the contributions of variables in two periods, i.e., the early 2010s (2010–2014) and the late 2010s (2015–2019); (3) discuss the policy implications for the current changes in ecotourism values and future forest protection and management.

2. Materials and methods

2.1. Study area

Lin'an County (29°56'N–30°28'N, 118°50'E–119°53'E), a suburb in the Hangzhou metropolitan area, is located in northwestern Zhejiang Province, China (Fig. 1). It covers 3118.8 km² with a population of about 0.54 million. The climate is a subtropical monsoon with an annual mean temperature of 16.9 °C and an annual precipitation of 1659.3 mm (LBS, 2020). The elevation varies between 5 and 1771 m. Lin'an is a typical mountainous area with forest coverage of 82%. Our object of this study is forest areas combined with the areas of other lands with tree cover (FAO, 2020), referred to as forests hereunder.

Lin'an has two national nature reserves (Tianmu Mountain, TM; Qingliangfeng, QLF), a national park (Fuchun River–Xin'an River, FXR), and a national forest park (Qingshan Lake, QSL). These areas were established in 1986, 1999, 1982, and 1999, respectively, and have corresponding tourist areas. They are rich in biodiversity with rare plants. However, local forestry productions, mainly bamboo shoots and hickory (*Carya cathayensis*) industries, with high economic value have been developed over the past few decades. Rural tourism of Lin'an first emerged in the TM region in 1994, and ecotourism with high socio-economic potential is booming. A concept, village scenic area, was first proposed in China by the local government in early 2017 to help develop the rural economy (LBS, 2018). Fifteen village scenic areas were established as pilot operations, six of which are currently priority areas for development (Fig. 1). During 2015 to 2019, the local government also focused on the construction of greenways at QSL forest park, where visitors can enjoy outdoor activities such as jogging and cycling to be closer to nature. Sustainable development of forests, therefore, requires a balance between ecosystem protection and socio-cultural benefits.

2.2. Methodology

The methodological framework included four main steps (Fig. 2). First, we extracted and classified some keywords from social comments of typical forests using natural language processing (NLP), and used the extracted keywords to develop explanatory public preference variables to present the contributions to forest ecotourism values. Second, we analyzed datasets acquired from earth observation, ground survey, and social media to map these explanatory variables (i.e., the identified preferences) at 30-m resolution for two periods: 2010–2014, 2015–2019. Third, the Maxent model was employed to assess forest ecotourism values using the locations of typical forests and the variables, evaluating the determinant variables and their contributions. Fourth, the changes in forest ecotourism values were identified by a hot spot analysis to explore the management and policy implications.

2.3. Social media data and text mining

Points of interest (POIs) from a popular open social media platform in China, Dianping, provide the location datasets related to ecotourism, such as scenic spots and homestay inns. Of all POIs that were within the study area before April 2021 via a public application programming interface (API) using Python, we collected longitude, latitude, description, and each comment: text and publication date. These datasets are publicly available, and the information protected by users' privacy was neither acquired nor analyzed. The publication date can speculate on the existences of POIs. All collected POIs were conducted to pre-processing which included coordinate transformation and data cleaning. The number of comments for each POI was counted for the two periods. One vital group of POIs was used as typical forests: scenic spots, such as nature reserves, beautiful villages, and botanical gardens. All POIs were carefully examined. If a POI was located in non-forest areas, we selected the nearest location in forests to replace it. In total, 100 valid POIs were finally stored in 2010–2014 and 103 valid POIs in 2015–2019. Another

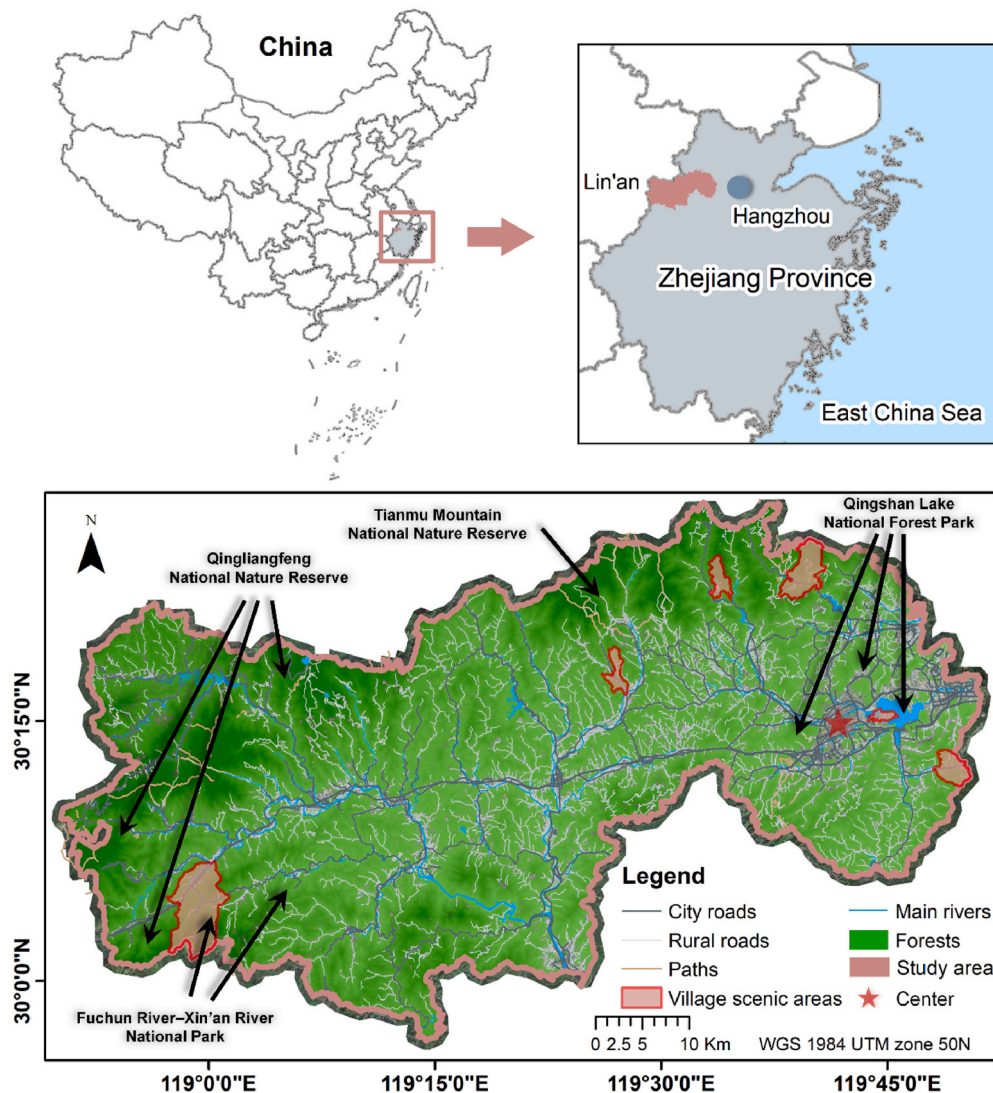


Fig. 1. Location of the study area (Lin'an, China). The forests were selected as the research object in the present study, and the figure also shows the distributions of the county center, the six prioritized village scenic areas and the main roads, paths and rivers.

group of POIs was inns, including agritainments and homestay inns. 18 and 238 POIs with non-zero comments were selected for the two periods of 2010–2014 and 2015–2019, respectively.

Text mining of comments is the key to exploring the relationship between public sentiments and preference variables to evaluate ecotourism values. Social comments include public preferences for why people traveled to the destination. A total of 6889 comments were collected from May 2006 to April 2021 for 103 scenic spots. Approximately 92% of comments were collected after 2015. The distributions of these comments by season, by year, and by area were all uneven, summarized in Supplementary Material.

Jieba NLP, an efficient library performing Chinese word segmentation, was used to extract words from the social comments. Some invalid words, such as destination, emoji, and conjunction, as well as words with fewer than twenty-five occurrences were discarded. Note that it was only counted once if a word (or a keyword below) occurred multiple times in a comment. The extracted words were manually organized into 25 words, referred to as the keywords, depending on their Chinese meanings. The text analysis was performed using Python, and the co-occurrence network of the keywords was visualized with Gephi 0.9.2 (Bastian et al., 2009) choosing the layout of Force Atlas.

2.4. Public preference variables

According to the keywords from text mining, the public preferences were categorized into the following types: accessibility, forest attribute, air environment, landscape heterogeneity, and color diversity. Combining the detailed description in comments, previous studies (e.g., Ciesielski and Stereńczak, 2021; Knoke et al., 2021; Vaz et al., 2019), and data availability, 27 variables were considered as candidates for the explanation of public preferences to forest ecotourism values (Table 1) and were computed for the two periods respectively. Variables were organized into two categories based on how they were mapped by the data sources: auxiliary GIS data (including the locations of inns from social media in Section 2.3) and RS data.

2.4.1. Data collection

For GIS-based variables, input data included the path networks in 2014 and 2019 (OpenStreetMap, 2020), the road and hydrographic networks, the locations of villages, the forest resource inventory (FRI) and the old trees census. Table 1 shows a part of detailed descriptions and methods for these variables.

For RS-based variables, 30-m Landsat 5/7/8 surface reflectance (SR) images (path/row: 119/039, 120/039) from January 2010 to December 2019 were used to generate the variables to describe the information on

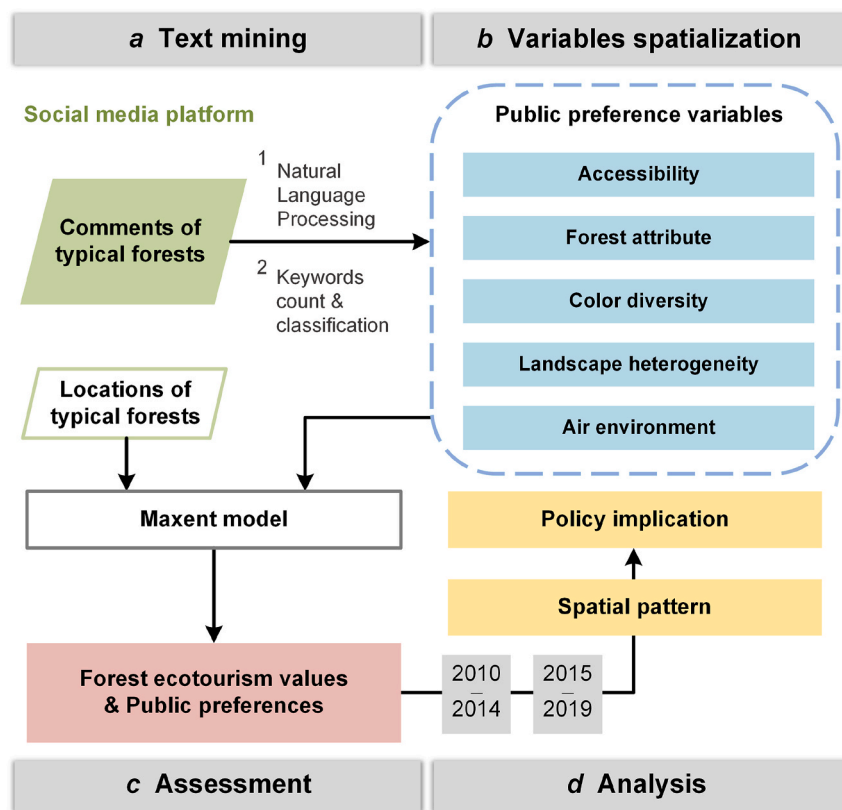


Fig. 2. Flowchart of the assessment and analysis for forest ecotourism values during the past decade: (a) text mining of social comments to propose a range of public preference variables; (b) spatialization for the variables acquired from earth observation, social media and ground survey for two periods; (c) Maxent models testing to assess forest ecotourism values and which set of variables best explained the contributions of public preferences; and (d) analyzing changes in forest ecotourism values.

forest attribute, color diversity, and landscape heterogeneity. Monthly temperature and annual PM_{2.5} concentration products were used to assess the variables of the air environment. 1-km monthly temperature data from 2010 to 2019 was downloaded from National Earth System Science Data Center (<http://www.geodata.cn>).

All datasets of the public preference variables were re-projected to WGS 1984, UTM 50N projection system with a spatial resolution of 30 m and were converted ASCII format files in ArcGIS 10.3 (ESRI Inc.). To avoid pair-wise correlations and multi-collinearity, the variables were performed Pearson correlation and variance inflation factors (VIF) analyses. We arranged several different combinations of variables with Pearson values < 0.6 and VIFs < 5 to further analysis (Menard, 2001).

2.4.2. Processing for remote sensing-based variables

2.4.2.1. Forest attribute. Trees and some specific tree species are widely mentioned in the social comments. According to the FRI of 2017 for the study area, the plantation areas of bamboo and hickory contributed approximately 37% of the entire vegetation coverage. The spatiotemporal distributions of the plantations were influenced by local policies and relevant industrial developments (You et al., 2020). Since the FRI data with the tree species level is not yet publicly accessible, we considered the land-use classes dominated by natural forests and plantations based on their forestry use properties. All available Landsat SR time series were selected and processed in GEE. Two land-use classes, i. e. natural forest and plantation, were generated by reclassifying the maps produced by You et al. (2020) for the two periods. The accuracies of classification results were measured using the F1-score values, which were all greater than 0.9 and met the needs of this study.

2.4.2.2. Color diversity. For characterizing color diversity, we used bands blue, green, red from the Landsat images to produce three variables. These bands are the visible part of the electromagnetic spectrum, related to the perception and experience of landscapes (Tveit et al.,

2006; Vaz et al., 2019). Multi-annual cloud-free image composites were generated for five-year intervals using a median reducer through GEE. We then calculated the standard deviation of the reflectance values of the pixels in the neighborhood (square with a side length of 210 m) based on the image composites for the three visible bands. Additionally, a one-term harmonic model (Eq. (1)) was applied to fit Landsat time series by a robust iteratively reweighted least squares method which used a cost function to reduce the influence of outliers during the fitting (O'Leary, 1990). The amplitudes of the generated time series represented the degree of changes within the period, describing the seasonal variability of reflectance values.

$$y(i, x) = a_i \sin(2\pi x/t) + b_i \cos(2\pi x/t) + c_i x + d_i \quad (1)$$

where $y(i, x)$ is the estimated value for the band i (Blue, Green, Red) at date x , a_i and b_i are the amplitude factors of seasonality for the band i , t is the number of days per year ($t = 365$), c_i is inter-annual changes for the band i , d_i is the overall reflectance value for the band i .

2.4.2.3. Landscape heterogeneity. Similar to Vaz et al. (2019), we applied a k-means clustering algorithm based on the previously generated image composites in GEE to describe landscape heterogeneity. A total of 20 clusters were obtained by training a sample of 30,000 pixels randomly for each composite. According to the clustered maps of the two periods, the following landscape metrics were calculated by moving window of square with a side length of 210 m through Fragstats software (version 4.2) (McGarigal et al., 2012): number of clusters, Shannon-diversity, Simpson-diversity, and Pielou evenness.

2.4.2.4. Air environment. To characterize the air environment, we performed two variables, annual mean temperature and annual mean PM_{2.5} concentration. The monthly temperature data was averaged to the five-year annual means for the two periods. The PM_{2.5} concentration annual 1-km datasets of 2013 and 2018 were generated following the methods described by Chen et al. (2021). The five-fold cross-validation results

Table 1
Descriptions of public preference variables considered as candidates.

Type	Variable	Description and method
Accessibility	CRoadDist	Euclidean distance to city roads
	RRoadDist	Euclidean distance to rural roads
	PathDist	Euclidean distance to paths
	MRivDist	Euclidean distance to main rivers
	Elevation	Elevation from a 30-m digital elevation model (DEM) acquired from Google Earth Engine
	Slope	Slope from a 30-m DEM
	Viewshed	Viewshed overlaps were calculated using the locations of all roads and paths, as well as DEM (Egarter Vigl et al., 2017; Yoshimura and Hiura, 2017)
Forest attribute	VillageDist	Euclidean distance to main villages
	InnDist	Euclidean distance to the locations of agritainment and homestay inns
	CanpDens	Tree canopy density
	OTreeDens	Old trees (estimated more than 100-year-old trees) density was calculated by the number of old trees within 1 km
	FUse	2-class forest land-use data, including natural forest and plantation
Color diversity	M_B; M_G; M_R	Median reflectance value for the band Blue/Green/Red, based on the multi-annual, cloud-free Landsat composites
	H_B; H_G; H_R	Spatial heterogeneity of median reflectance values for the band Blue/Green/Red
	Var_B; Var_G; Var_R	Seasonal variability of reflectance values for the band Blue/Green/Red based on the multi-annual, cloud-free Landsat time series
Landscape heterogeneity	Clust_n; Clust_shdi; Clust_sidi; Clust_pie	Number of clusters; Shannon-diversity, Simpson-diversity and Pielou evenness of clusters
Air environment	AMT	Annual mean temperature (°C)
	PM _{2.5}	Annual mean PM _{2.5} concentration (μg/m ³)

showed that R^2 values were greater than 0.85 and met the needs of this study.

2.5. Assessing forest ecotourism values and the contributions of public preferences

In this study, we used Maxent modeling software (version 3.4.4) (Phillips et al., 2018) to model and map the forest ecotourism values. Maxent uses a machine-learning algorithm to predict the probability of presence of a study object through the relationship between its given observed points and a set of potential explanatory variables, such as environmental and social factors (Phillips and Dudík, 2008). The observed points are referred to as presence samples. It is convenient that only the variables and the presence samples are required to be put into Maxent while the absence samples are not. Maxent is robust against even limited samples and, hence, is widely applied to predict species niches and distributions (Phillips et al., 2006; Phillips and Dudík, 2008). Currently, Maxent also showed the advantages of socio-cultural value mappings in the natural and semi-natural ecosystems (He et al., 2019; Sherrouse et al., 2014; Yoshimura and Hiura, 2017). The area under the curve (AUC) of the receiver operating characteristic (ROC) plot is employed to assess the performance of the model. The ROC curve is plotted by 1 -specificity (as the abscissa) and sensitivity (as the ordinate). 1 -specificity represents the false positive rate and sensitivity represents the true positive rate (Fielding and Bell, 1997). ROC can evaluate the efficiency of a binary classifier, e.g., Maxent, and AUC implies the probability of sensitivity. ROC is calculated through thresholds that relied on the predicted results in a Maxent model and thus AUC can be used for the comparison of different models. An AUC

value ranges from 0 to 1, which indicates a random model at 0.5. The AUC value presents the model performance, the higher the better. AUC values of 0.5–0.7, 0.7–0.9, and > 0.9 suggest a model with moderate accuracy, good accuracy, and high accuracy, respectively (Swets, 1988).

Our ecotourism value maps were the probability of presence of typical forests, presenting the potential ecotourism areas. The locations of scenic spots as presence samples and the spatialized public preference variables were integrated into Maxent. These POIs are high representative and diverse for forest ecotourism in the study area, and some of the places with fine values of ecotourism services may be paid little attention to be recorded in the Dianping platform yet. Consequently, Maxent is suitable for analyzing these POIs. 90% of POIs were randomly selected for training and the rest were used for validation. The procedure was repeated 10 times for cross-validation to eliminate randomness and then the predicted results were averaged as final outputs.

Concurrently, Maxent estimated relative contributions of input variables using the jackknife test, as well as response curves. To estimate the contributions of different variables, the contribution of a given variable was added when the regularized gain increased and subtracted when the regularized gain decreased during the iterative training. Furthermore, the response curves can show how the AUC changes as each variable was varied, using the Maxent model created with only the corresponding variable. In the course of the initial modeling process, a variable would be considered to discard if the corresponding response curve changed slightly as the variable varied.

2.6. Spatiotemporal analysis on forest ecotourism values

We divided the values of forest ecotourism into three classes (high, medium, and low) using the geometrical interval classification for spatial visualization and area statistics in the two periods. For comparative analysis, the classification standard of the values in 2010–2014 was aligned with that in 2015–2019.

A hotspot analysis using *Getis-Ord Gi** statistics was employed to identify the spatial patterns of statistically significant hot spots (high values) and cold spots (low values) (Getis and Ord, 1992) based on the observed changes of forest ecotourism value over the two periods. Five clusters were generated by the resultant Z-Scores and *p*-values of 90% and 99% confidence intervals, namely hot-spot area, second hot-spot area, stable area, cold-spot area, and second cold-spot area. The resultant map is referred to as a spatial pattern map.

3. Results

3.1. Public preferences of forest ecotourism values

Fig. 3 provides an insight into the co-occurrence relations of the 25 keywords. One keyword (likes) is related to the public sentiment, and the rest of the keywords are related to the public preferences. About 66% of the social comments directly mentioned that visitors liked these typical forests, and half of the comments referred to information about the landscape. A notable proportion of the attraction to visitors was accessibility-related information: paths (48%), roads (37%), rivers (26%), cableways (17%), and elevation (15%), for example. Visitors also had certain demands on the air environment, preferring fresh air (18%), comfortable temperatures (14%), good weather (13%), and the quietness of forests (4%). For forest ecotourism, trees (14%) were undoubtedly one of the subjects of interest, especially some specific tree species (13%) and old trees (6%). The keywords as nodes are presented by circles and the edges show the co-occurrence relationships of pairs (Fig. 3). Besides, the more frequently keywords occurred together in the comments, the closer the spatial distribution of them is in the figure. For example, the keywords that co-occurred most frequently were “likes-landscape” (2555), “likes-path” (2243), and “landscape-path” (1803), indicating that the visitors prefer the landscape along the path.

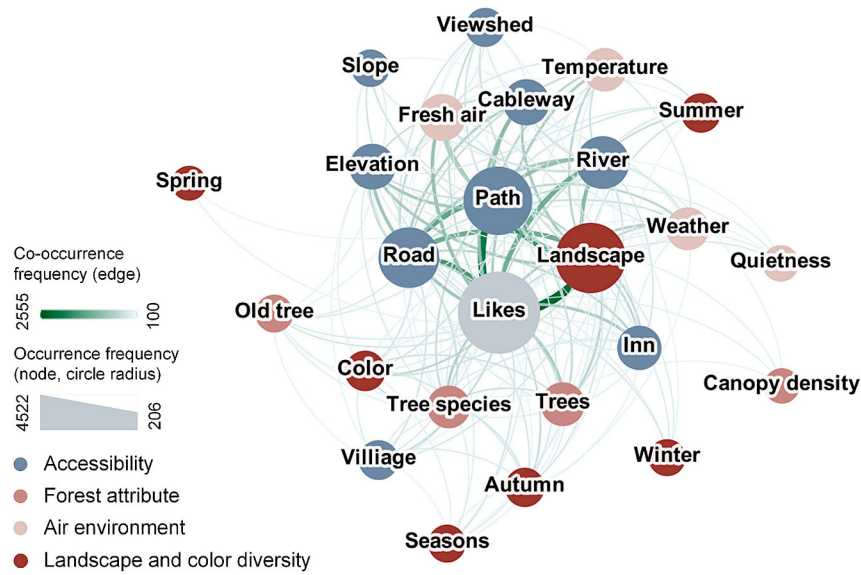


Fig. 3. The co-occurrence network based on the text mining results to indicate the locally relevant public preferences in the study area.

3.2. Spatiotemporal patterns of forest ecotourism values

Considering the AUC values and the corresponding standard deviations, we separately selected the prediction results of the optimum model in the two periods for further analysis (see Supplementary Material for other prediction results). The AUC values of the two models were greater than 0.8: 0.885 (± 0.061) for the early 2010s and 0.894

(± 0.041) for the late 2010s respectively. The results suggest that ecotourism value mappings using Maxent were satisfactory in terms of accuracy and robustness. The two ecotourism value maps were comparable and can reveal the dynamics of forest ecotourism value in the study area over the past decade.

Fig. 4 presents the spatial distributions of forest ecotourism values in the two periods. These two ecotourism value maps are visually similar as

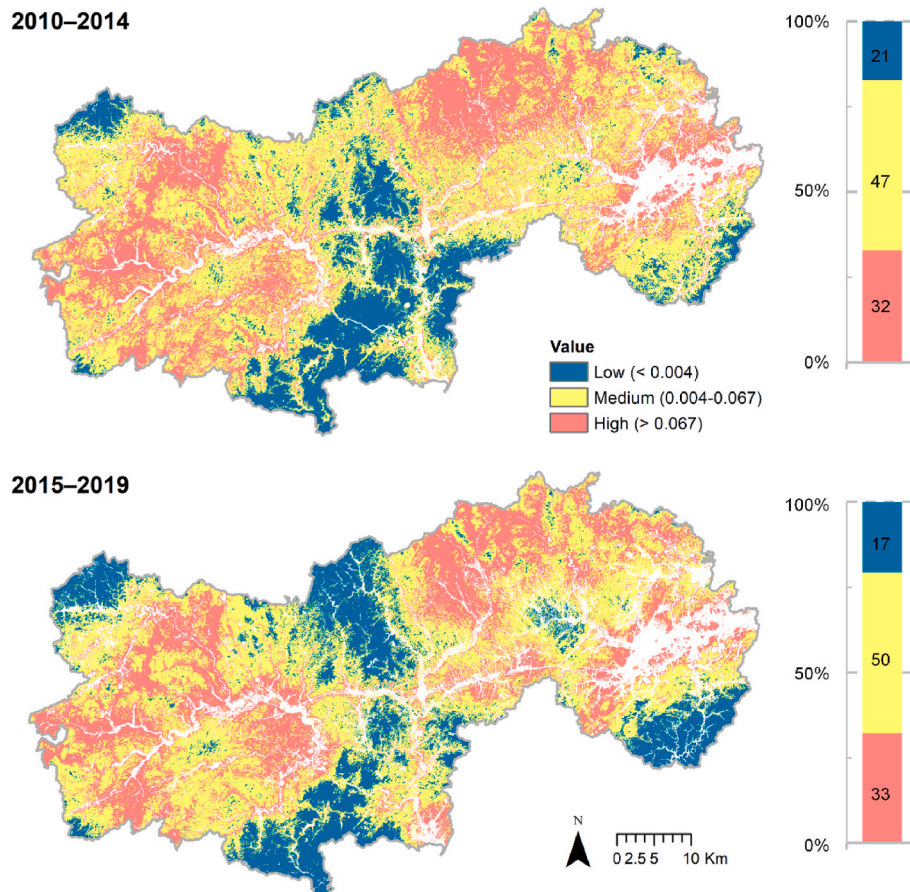


Fig. 4. The spatial distributions of forest ecotourism values generated for the two periods, respectively.

a whole, but there are some discrepancies in parts. In the early 2010s, high-value forests were located mainly in three clusters, in the north-eastern, western and eastern Lin'an. The northeastern cluster widely covered TM nature reserve. The western cluster included QLF nature reserve and FXR park. The eastern cluster mainly covered the county center and QSL forest park. By contrast, during 2015–2019, high-value forests were relatively scattered with some new areas of high ecotourism value near towns, such as central and southern Lin'an. A significant increase in low-value forests was observed in northern and southeastern Lin'an. Besides, an inspection of the area proportions at each class showed a minor change in the overall forest ecotourism value throughout the study period (Fig. 4). The increases of low-value forests (4%) from 2010 to 2019 came at the expense of decreases in high-value forests (1%) and medium-value forests (3%).

Despite the minor net change, a considerable change in forest ecotourism values was observed from 2010 to 2019 (Fig. 5). A significant number of areas experienced a decrease in forest ecotourism values (cold-spot areas, 44%), and meanwhile, substantial areas (34%) were hot-spot areas of ecotourism value changes. Moreover, the second hot-spot area, stable area, and second cold-spot area accounted for 3%, 16%, and 3%, respectively. Many hot-spot areas for ecotourism value changes were along the main city roads passing through the county from east to west. In terms of the national scenic spots and surroundings, the TM region, where rural tourism first emerged, were mainly covered cold-spot areas, while the QSL regions were occupied a large number of hot-spot areas. In addition, the changes in forest ecotourism values were visually inconsistent among the six prioritized village scenic areas. Further statistics on these village scenic areas showed that approximately three quarters of the areas they covered were the hot-spot and second hot-spot areas and 10% were stable areas. Overall, the percentages of the cold-spot and second cold-spot areas are insignificant.

3.3. Contributions of public preference variables

Table 2 shows the contributions of the variables based on the public preferences for modelling the forest ecotourism values using Maxent for the two periods. Among the types of variables, critical percentages of contributions came from the variables related to accessibility for both periods (67% in 2010–2014; 72% in 2015–2019), followed by the variables of forest attribute (11% in 2010–2014; 13% in 2015–2019) and color diversity (17% in 2010–2014; 11% in 2015–2019). In contrast,

Table 2

The contributions of used variables for the ecotourism value maps of forests for the two periods.

Type	Variable	Contribution (%)	
		2010–2014	2015–2019
Accessibility	PathDist	29.7	27.3
	InnDist	19.1	26.1
	CRoadDist	10	9.9
	RRoadDist	3.1	3.4
	Elevation	2.4	3.1
	Viewshed	2.4	2.1
Forest attribute	FUse	8.1	7.7
	OTreeDens	2.1	3.3
	CanpDens	0.9	2
	H_G	13.1	8.2
Color diversity	Var_G	0.6	1.6
	Var_R	3.1	1.1
	Clus_sidi	4.1	1
Landscape heterogeneity	PM _{2.5}	1.2	3.1

landscape heterogeneity and air environment accounted for less than 5%. In the context of specific variables, note that PathDist and InnDist acquired from social media showed excellent contributions to predict forest ecotourism values in the two periods. The top four variables (PathDist, InnDist, H_G, and CRoadDist) contributed approximately 72% of the prediction models during 2010–2019.

The influence patterns of each variable were measured by response curves (Fig. 6). Mostly, the effects of the variables were similar for the two periods. The public preferred the forests, which are accessible by paths, with broad viewshed and convenient transportation. Within a certain range, forest ecotourism values were negatively correlated with road distances (approximately < 1 km from city roads; < 700 m from rural roads). Both high and low elevation areas were popular. In addition, old trees, high color diversity, fresh air contributed greatly to the increase of forest ecotourism values. Natural forests paid more attention than plantations. Open spaces and dense forests were both preferred by people. These influence patterns explained the diverse effects of different variables on ecotourism value estimates in the study area.

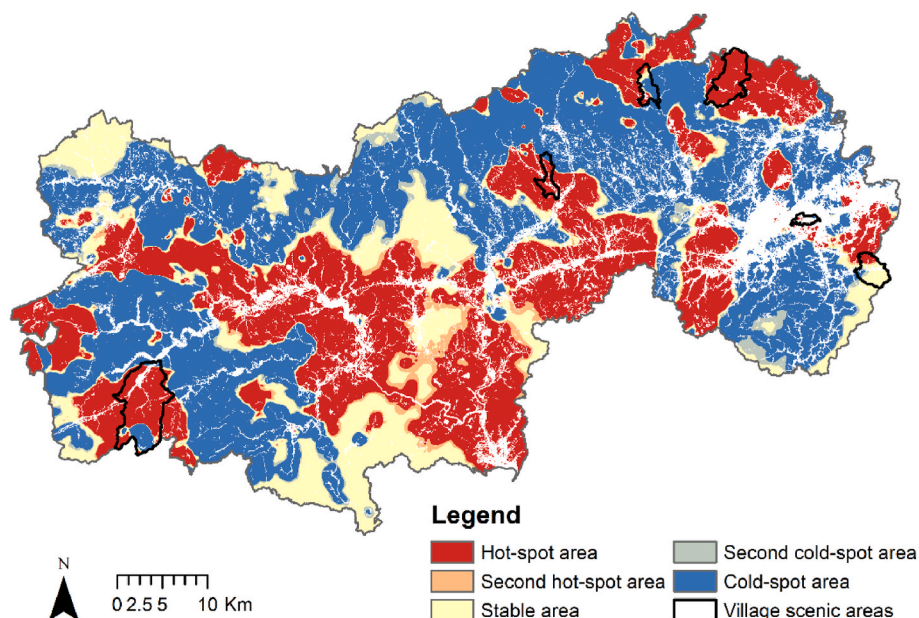


Fig. 5. The spatial pattern of changes in forest ecotourism values during 2010–2019.

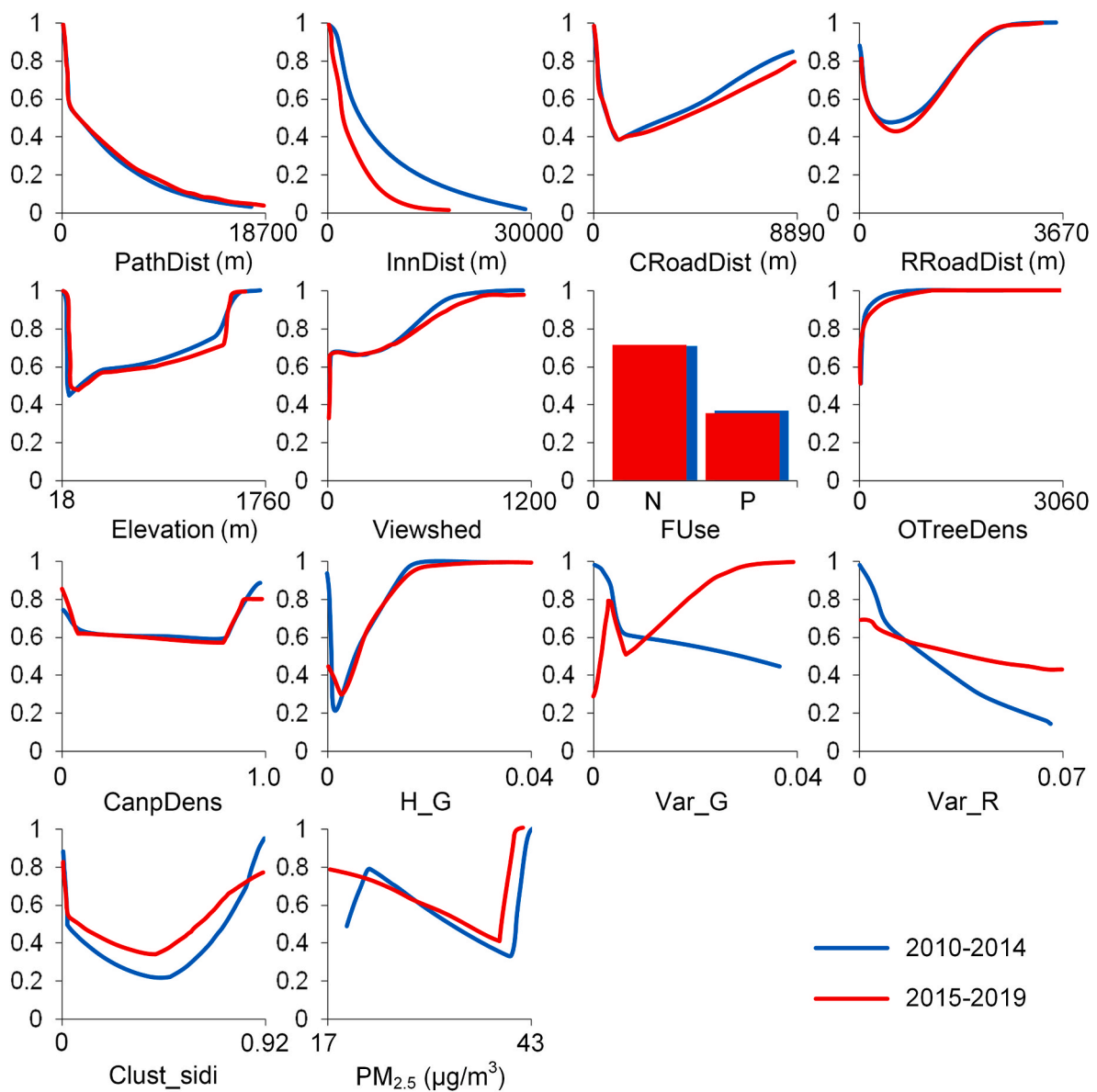


Fig. 6. Response curves of each variable on ecotourism value mappings of forests in the two periods. N: natural forest; P: plantation.

4. Discussion

4.1. Public preferences and multi-source data

Information on public preferences is essential for making targeted forest management guidelines, which can effectively integrate socio-cultural benefits into conservation policies and support ecotourism planning and development (Edwards et al., 2012; Plieninger et al., 2015; Schirpke et al., 2021). The assessment of forest ecotourism values on a regional level still lacks an appropriate methodology covering the spatiotemporal dimension, which limits our ability to understand their spatial and temporal dynamic properties and to manage forests in a way as to meet the needs of inhabitants and visitors. In contrast to other studies which assessed recreational or aesthetic values (e.g., Schirpke et al., 2019; Van Zanten et al., 2016; Vaz et al., 2019), our proposed methodology includes people's preferences. We also found that the actual use of forests for ecotourism is a result of varied landscape preferences and context variables (e.g., accessibility). Concurrently, multi-source data helps to identify these ecotourism values and public preferences.

Quantitative information on public preferences can increase efficiencies in the uses of forest resources. The contributions of public preferences to ecotourism value were significantly related to the variables of accessibility acquired from GIS data, especially PathDist and InnDist both from citizen participation platforms. Our findings suggest that managers can regulate the distribution of forest ecotourism values by changing accessibility through, for example, adding/removing the construction of transportation infrastructure and controlling the distribution of inns. The Maxent analysis also showed the certain contributions from forest attribute and air environment to ecotourism value assessment in the context of forests. These variables were rarely involved in explaining recreation and aesthetics (Felipe-Lucia et al., 2018; Knoke et al., 2021) because they were considered to be within the scope of provisioning and regulating services in present studies. In fact, the freshness of the air, the height of temperature, and the shape of the trees are all related to the human perception of outdoor activities in forests (Knoke et al., 2021; Wang et al., 2020). Besides, the public preferences for forest attribute such as forest types and tree species are inconsistent among regions (Edwards et al., 2012). These suggest strengthening the protection and management of special environments

such as old trees and villages to meet the needs of visitors. Thus, the knowledge on public preferences obtained from social media efficiently supports the construction of an objective and local indicator system for assessments. Satellite data has also facilitated these assessments by capturing spatiotemporal information about the visual and sensory attributes of ecosystems, such as seasonal features of landscapes. The results of the text mining reveal various correlations among the keywords, suggesting that there are differences in public preferences for seasons (Fig. 3). Public preferences in autumn were related to the changing colors of trees, and those in summer were associated with rivers and temperatures. It suggests that managers need to have appropriate ecotourism management and planning for different seasons in the same area to meet visitors' demands (Gosal et al., 2019; Schirpke et al., 2018).

4.2. Policy implications from forest ecotourism value dynamics

In the context of land use change and climate anomalies, forest ecotourism values will be more affected in the future (Inglis and Vukomanovic, 2020; Sannigrahi et al., 2019). The increase in green space and the length of paths in low-altitude areas will reduce outdoor activities and aesthetic experiences in some high-altitude areas (Schirpke et al., 2019). Besides, people are attracted to cooler mountain areas in summer due to increased temperatures. However, reduced snow cover due to climate change will result in fewer areas of high ecotourism values for outdoor activities such as skiing.

The spatial pattern of changes in forest ecotourism values reveals the transformation of the potential for local ecotourism. Our results show that hot-spot areas were mainly observed in the central areas of Lin'an, where there are densely distributed towns and villages. Concurrently, forest ecotourism values have increased in the village scenic areas, QSL forest park, and their surroundings, which the local government focused on in recent years. Local government and managers improved rural environment and infrastructure, and integrated local folklore, cuisine and other characteristic resources to build each village scenic area. Rural tourism, which focuses on the experiences of nature and country life, is being promoted vigorously in here. Fifteen new village scenic areas received a total of 805,200 visitors during 2017–2019. Local village communities and villagers increased their incomes by 19.62 million yuan and 1.85 million yuan, respectively. Another priority is that a project for comprehensive management and protection has been carried out around QSL forest park. The greenways, up to 42 km in length, were built along the lake between 2015 and 2019, providing recreational and aesthetic opportunities. In November 2019, the subway between Lin'an and Hangzhou city opened, improving the accessibility of Lin'an, especially the QSL regions. The demand for forest ecotourism will probably further increase. In addition, although high-value forests were distributed throughout the TM region in both periods (Fig. 4), forest ecotourism values were identified through the spatial pattern map (Fig. 5) as decreasing within TM nature reserve and increasing within the nearby village scenic area. It implies that ecotourism services are mobile when the demand for ecotourism is in a stable state. There are competitive relationships among ecotourism potential areas in the same region. These observations indicate that local policies promote the flow of ecotourism services and the distribution of ecotourism resources.

Our spatial pattern map clearly presents a considerable ecotourism value change over the past decade and can identify priority areas for forest resource management and biodiversity conservation. Considering the impacts of changes in forest ecotourism values on key biodiversity areas, we conducted further zonal statistics on changes within the two national nature reserves. A nature reserve is generally managed in three parts in China: core areas, buffer areas, and experimental areas (where moderate ecotourism activities are permitted). According to our statistics, 30% (39%) of the core areas (buffer areas) were affected by the hot-spot and second hot-spot areas, 48% (45%) of which were occupied by the cold-spot and second cold-spot areas. These findings suggest that overall management pressures have dropped, but noteworthy

management areas may need to be adjusted accordingly. Additionally, the cold-spot areas in the north-central and southeastern areas of Lin'an covered the newly-planted hickory areas and long-term planted bamboo areas, respectively (Fig. 5). Few understory plants grow in the hickory areas, and the degradation of bamboos leads to color changes. It is foreseeable that the aesthetic values in these areas are weakened. It implies that we should find a two-pronged strategy to reduce the trade-offs between ecosystem services (Zheng et al., 2019).

4.3. Methodology considerations and future needs

The public preferences from social media and earth observation offer some new perspectives to support the assessment of forest ecotourism values. Remotely sensed images can provide spatiotemporal physical information of earth surface and are widely used to describe biodiversity variables to support ecosystem services assessments (Cord et al., 2017). However, studies on cultural ecosystem services are required to integrate remote sensing data with social media data (e.g., Van Zanten et al., 2016; Vaz et al., 2019). Social media data includes information, such as human emotions (Kim et al., 2019) and land use (Huang et al., 2020), that is not available from remotely sensed images. It promotes our ability to understand human-environment interactions and to analyze how people sense and change their living environment (Zhu et al., 2019).

There are still several challenges. First, unstructured social media data is without an appropriate organization which cannot be directly linked to remote sensing images (Zhu et al., 2019). To explore public preference variables, we used social comments on typical local forests by an NLP method. Recently, some studies combined NLP methods with photo analysis to provide more detailed mining of the variables for assessing recreational and aesthetic values (Gosal et al., 2019; Gosal and Ziv, 2020). Suitable NLP methods can be used in the future to mine rich textual information to analyze public perception.

Second, social media data can be biased and has ethical concerns, which need to be considered in future studies. The representativeness of data from social media has been discussed critically for different platforms attracting specific groups of users (Minin et al., 2015). Our data only came from Dianping users, mainly young people, and the visit frequency in mountainous areas is not as high as that in cities. But comments and POIs from social media have avoided some shortcomings of traditional methods for data collection. Especially in some areas where resources for ground-based data collection are scarce, social media data may help save costs. Further integrating different social media platforms and analyzing different or unique beneficiary groups separately can suggest more detailed and dynamic strategies to manage ecotourism areas (Gosal et al., 2019). Because of the late popularity of social media, early data is relatively lacking. Although the number of inns used in the two periods was quite different, the distribution areas of inns were consistent with the real situation in consultation with experts. Notably, there are also ethically sensitive issues with social media data, such as users' privacy and free speech, which should be carefully addressed before using it for sustainability and other studies (Minin et al., 2015).

Third, at present, RS data still has great potential for mining information of visual perception (Cord et al., 2017). Few indicators were available to describe color diversity and landscape heterogeneity (Cieśielski and Stereńczak, 2021; Vaz et al., 2019). The indicators were explored and played a role in prediction models in this study. Besides, there are fewer studies on cultural ecosystem services in different months, seasons, etc. (Vaz et al., 2019). It is necessary to assess the seasonal values of ecotourism and other services in designated areas to meet more detailed forest resource management and ecotourism planning in the future.

We considered 27 variables to assess the forest ecotourism values based on the keywords from text mining and detailed descriptions of comments. Due to the limitations of data collection, some keywords

(such as cableway) and some high-frequency human activities (such as rafting, skiing) were discarded in the subsequent model predictions. Further assessments could consider more influencing factors in terms of the practice of sports (such as adventure tourism) to promote the recovery of degraded areas and the conservation of sensitive areas. The calculation of the viewshed did not consider the routes of these activities and the height of the trees (Yoshimura and Hiura, 2017). In addition, the use of satellite data with different spatial resolutions may have some impacts; for instance, the 1000-m air environment data was resampled to 30-m. Despite these limitations, high accuracies were achieved in our ecotourism value maps. The adopted methodology has the potential to be extended to the assessment of cultural services for other mountainous areas or rural areas in different ecosystems by incorporating and adjusting the corresponding public preference variables into the prediction model.

5. Conclusions

Assessment of forest ecotourism values is crucial to understanding the interactions between public preferences and ecosystems, and providing relevant information for sustainable ecosystem management and ecotourism development. In this study, we performed a feasible framework to identify changes in forest ecotourism values based on public preferences using Maxent models and multi-source data. We found that social comments and remote sensing data can provide objective descriptions of public preferences. Forest ecotourism values only showed a net decrease of 1–3% over the past decade in Lin'an. But a considerable increase was observed in 34% of forests, concentrated in the village scenic areas, QSL forest park, and their surroundings. Besides, 44% of forests experienced a significant decrease in ecotourism values, mainly in some planted areas. The contribution of public preferences to ecotourism value revealed that changes in accessibility variables were the major attribution of such changes, while the variables related to forest attribute and color diversity also played a part. Such information can improve the efficiency of forest resource use. Our findings imply that local rural-building policies have influenced the distribution and intensity of forest ecotourism. With the increasing pressure on sustainable development and rural revitalization, our proposed methods and findings can assist decision makers to evaluate the response and effectiveness of local policies, and make proactive efforts for forest protection, restoration, and management.

CRedit authorship contribution statement

Shixue You: Conceptualization, Methodology, Data curation, Software, Visualization, Writing – original draft, Writing – review & editing. **Qiming Zheng:** Conceptualization, Writing – original draft. **Binjie Chen:** Data curation, Software, Validation, Writing – review & editing. **Zhihong Xu:** Investigation, Resources. **Yue Lin:** Visualization, Writing – review & editing. **Muye Gan:** Formal analysis, Writing – review & editing. **Congmou Zhu:** Writing – review & editing. **Jinsong Deng:** Formal analysis, Funding acquisition, Supervision. **Ke Wang:** Project administration, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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